



DEPARTMENT OF ENGINEERING CYBERNETICS

TTK4900

MASTER THESIS

Adaptive Filtering of Blood Velocity Signals During Running

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Abstract

Acknowledgments

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Introduction

Chapter 2

Theoretical background

2.1 Ultrasound

Ultrasound is defined as sound waves with frequencies above the upper limit of human hearing, which is typically around 20 kHz.

2.1.1 Transducer

2.1.2 Reflection and Scattering

2.1.3 Attenuation

2.1.4 Time of Flight

2.1.5 Noise sources

2.2 Pulsed-Wave Doppler

In the present system, PW Doppler is used to measure the blood flow velocity. It works by emitting a series of high frequency sound pulses into the tissue at a constant Pulse Repetition Frequency (PRF). Each pulse consists of a burst of sound waves with a pulse center frequency f_c with N_{cycles} cycles. The pulse length T is then given by $T = N_{cycles}/f_c$. After a pulse is emitted, the system records the returning echoes. This is commonly referred to as the *fast time* signal. The fast time signal is sampled with a sampling rate $f_{s,FT}$

2.2.1 Decimation

2.2.2 Clutter Filter

Parameter	Symbol	Value
Pulse center frequency	f_c	8 MHz
Number of cycles	N_{cycles}	12
Pulse length	T	1.5 μs
Pulse repetition frequency	PRF	20 kHz
Speed of sound in tissue	c	1540 m s ⁻¹

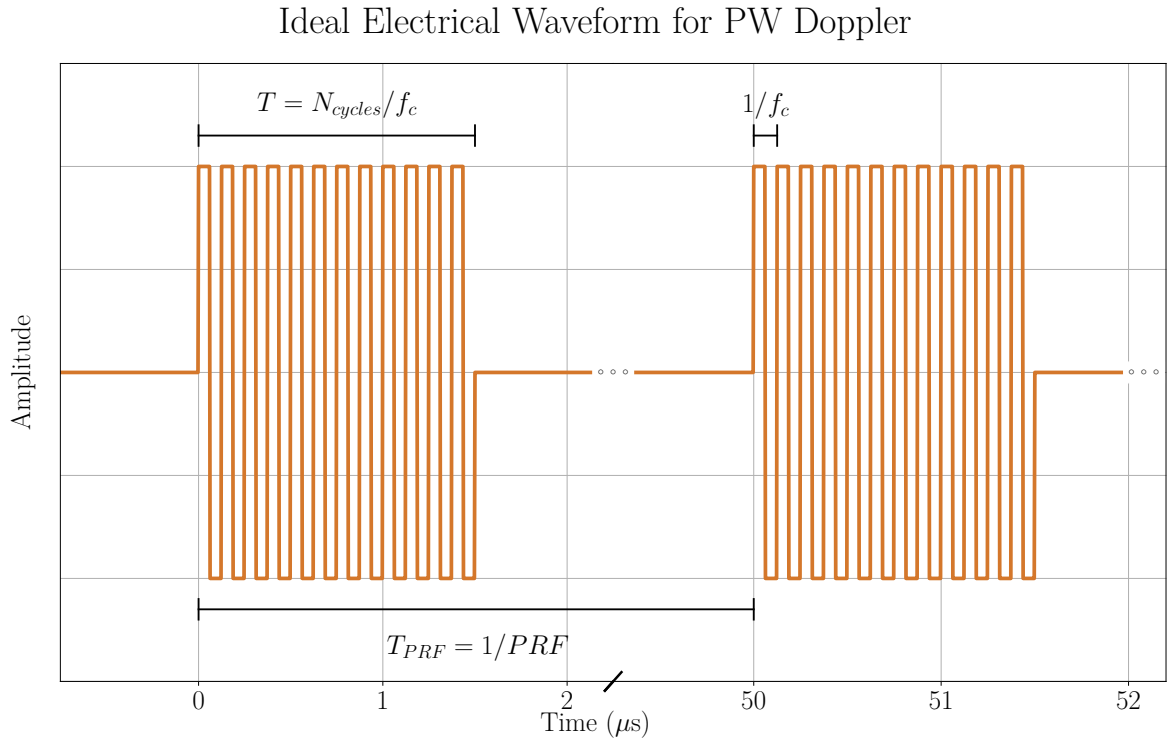


Figure 2.1: Ideal emitted electrical waveform for PW Doppler, with annotations. Note that the time axis is broken to show the pulse repetition period.

Chapter 3

Method

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion